Photonic Substrate Intelligence

A Unified Architecture for Speed-of-Light Neural Network Inference, Wavelength-Multiplexed Massive Parallelism, and Holographic On-Device Training via Thermomagnetic Physical Substrates

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Abstract

We describe a unified speculative architecture for neural network computation in which inference, massive parallelism, and on-device training are performed primarily in the optical domain, with two distinct classes of physical substrate serving complementary roles: a write-once DNA cassette tape providing petabyte-scale context input, and a rewritable sputtered magneto-optical thin-film tape serving as weight store, training target, and gradient-encoded write medium. The architecture proposes to eliminate digital conversion from the critical computation path entirely, replacing electronic matrix multiplication, nonlinear activation, and gradient descent with their optical and thermomagnetic equivalents.

The system comprises seven interlocking components: (1) a two-substrate physical input architecture — DNA cassette tape optically imaged as context input, and magneto-optical thin-film tape storing reconfigurable weight patterns; (2) reconfigurable Spatial Light Modulator arrays implementing ternary-optimised weight multiplication layers; (3) passive nonlinear optical activation elements — either volumetric metamaterial photonic crystals or fibre-optic saturable absorbers — implementing intensity-threshold activation without electronic conversion; (4) a wavelength-division multiplexed optical inference stack supporting N concurrent independent model forward passes through a single physical column of optical elements; (5) a volumetric array of such stacks — specifically, a 100x100 planar array forming a 30-centimetre cube — providing 10,000xN concurrent frontier-class model inferences at kilowatt-scale power; (6) an Orbital Angular Momentum-encoded gradient write beam that encodes the layer-wise gradient tensor as structured light and directs it to the weight substrate; and (7) thermomagnetic domain writing via focused laser, updating weight patterns on the magneto-optical thin-film tape in response to the gradient-encoded beam, closing a fully photonic training loop.

The combined architecture suggests a physically compact, low-power, network-isolated intelligence substrate capable of running and training tens of thousands of concurrent frontier-class neural network instances in a form factor compatible with vehicle or secure-room deployment. All component concepts draw upon established physical principles, prior experimental demonstrations, and — in the case of the DNA context substrate — hardware now physically realised. The novel claim is architectural combination. This document constitutes a prior art disclosure establishing date of conception as of 13 March 2026. Experimental validation is required at every stage and is not claimed.

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1. Introduction

Every neural network inference that has ever been performed — every large language model response, every image synthesis, every protein structure prediction — has run on a variant of the same fundamental computing architecture proposed by von Neumann in 1945: a processor separated from memory by a data bus, with computation performed by switching transistors clocked at discrete intervals. The history of AI hardware is a history of optimising within this architecture: faster clocks, wider buses, more parallelism across transistor arrays, closer memory integration. Each advance reduces the cost of computation within the paradigm. None escapes the paradigm.

The paradigm has well-characterised ceilings. Memory bandwidth limits how fast data can be moved between storage and compute. Transistor switching generates heat proportional to frequency and count. Clock speeds have plateaued physically at a few gigahertz. Power consumption scales with compute demand. At frontier model scale, these limits manifest concretely: a single H100 GPU draws approximately 700 watts; running 10,000 concurrent frontier-class model instances requires on the order of 10,000 such GPUs, implying a capital cost exceeding £300 million, a power draw of approximately 7 megawatts, and physical infrastructure requiring purpose-built data centre facilities with dedicated grid connections.

This paper does not propose to optimise within that paradigm. It proposes a different physical substrate for the inference and training operations of neural computation: light propagating through engineered optical elements, with physical magnetic media serving as the memory that light reads and rewrites. The motivation is not merely speed — though the architecture suggests sub-nanosecond inference latency — but a qualitative inversion of the cost structure of large-scale AI deployment: from a regime where additional inference capacity requires proportional additional power and capital, to one where inference capacity scales with optical path replication at fabrication cost, and the marginal cost of an additional concurrent inference channel approaches zero.

The architecture draws upon and integrates seven established research areas: diffractive optical neural networks (Lin et al., 2018); photonic matrix multiplication via spatial light modulators (Lightmatter; MIT; Stanford); ternary-weight neural networks (Microsoft BitNet b1.58, 2024); orbital angular momentum multiplexing in optical communications (Wang et al., 2012; Bozinovic et al., 2013); thermomagnetic domain writing in magneto-optical recording; holographic associative memory and optical backpropagation (Gabor, 1969; Hughes et al., 2019); and high-density DNA information storage on physical cassette media (Jiang et al., 2025). The novel contribution is the specific architectural combination of these fields into a unified inference-and-training system, not the invention of any individual component. Prior art in each subfield is explicitly acknowledged throughout.

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2. Background and Prior Art

2.1 Diffractive Optical Neural Networks

Lin et al. (2018) demonstrated that passive diffractive surfaces, computationally trained and fabricated by 3D printing, could perform multi-layer optical classification at the speed of light without electronic computation. Each diffractive layer modulates the phase and amplitude of incident light, implementing a learned linear transform; cascaded layers perform deep inference. This foundational result establishes the physical viability of passive optical inference and directly informs the activation layer design proposed here. Its primary limitation — that trained weights are fixed at fabrication and cannot be updated without printing new layers — is addressed by the reconfigurable weight layer architecture of Section 3.2 and the thermomagnetic training loop of Section 3.6.

2.2 Optical Matrix Multiplication via Spatial Light Modulators

Spatial Light Modulators are commercially mature devices modulating incident light on a per-pixel basis at refresh rates of tens to hundreds of milliseconds. Their use for reconfigurable optical matrix multiplication is established in the photonic computing literature and commercialised by Lightmatter (Boston, MA), with research implementations at MIT and Stanford. SLMs implement mutable weight layers — in contrast to the fixed fabricated surfaces of D2NN — making them the natural component for updateable optical neural networks. The present architecture exploits the ternary weight constraint (Section 3.2) to reduce SLM configuration complexity substantially relative to prior continuous-gradient approaches.

2.3 Ternary-Weight Neural Networks

Microsoft Research's BitNet b1.58 (Ma et al., 2024) demonstrated that large language models can be trained with weights constrained to three values — {-1, 0, +1} — with competitive performance at equivalent scale. This constraint is significant for the present architecture because ternary weights map directly to three discrete SLM pixel states: phase-invert, block, and transmit respectively. This reduces SLM configuration from continuous analogue gradient to three discrete states per pixel, substantially simplifying fabrication tolerances, reconfiguration logic, and OAM gradient encoding for training.

2.4 Wavelength-Division Multiplexing

Wavelength-division multiplexing exploits the physical independence of different wavelengths of light in linear optical media: photons at lambda_1 and lambda_2 do not interact and carry fully independent information. Dense WDM systems routinely carry 80 or more independent channels on a single physical medium. Photonic crystal structures are inherently wavelength-selective — bandgap frequencies are determined by periodic geometry — enabling independent optical responses at distinct wavelengths within the same physical element. The application of WDM channel independence to concurrent multi-model neural network forward passes through a single optical stack is proposed here as a novel combination not present in the optical computing literature.

2.5 Orbital Angular Momentum of Light

Light beams carry angular momentum in two forms: spin angular momentum associated with circular polarisation (+/-h_bar per photon), and orbital angular momentum (OAM) associated with helical phase fronts characterised by integer topological charge l, where l is an element of the integers and is unbounded. Beams at different OAM modes are fully orthogonal in free space and separable by mode sorters. OAM multiplexing has been demonstrated at terabit-per-second aggregate data rates (Wang et al., 2012; Bozinovic et al., 2013). The present architecture proposes the first application of OAM mode superposition to the encoding of neural network gradient tensors as structured light for physical weight update.

2.6 Holographic Associative Memory and Optical Backpropagation

Gabor (1969) proposed that holographic interference implements associative memory: the interference pattern between a reference beam and an object beam encodes their relationship as a physical hologram, such that presenting one recalls the other. This outer-product structure is mathematically equivalent to a component of the backpropagation gradient. Hughes et al. (2019) demonstrated that optical backpropagation is physically realisable in a photonic stack: in-situ gradient measurements via phase conjugation can drive weight updates without digital conversion. The present architecture extends this to gradient encoding on a write beam directed at a physical magnetic substrate external to the inference stack.

2.7 Magneto-Optical Recording and Thermomagnetic Writing

Magneto-optical recording exploits the thermomagnetic effect: a focused laser beam heats a magnetic medium above its Curie temperature, reducing coercivity to near zero, allowing a bias field to write a domain of chosen polarity on cooling. This mechanism is commercially demonstrated in MO discs (rated approximately 10^6 rewrite cycles) and in heat-assisted magnetic recording (HAMR) for hard disc drives, where sub-diffraction-limit domain writing is achieved via precise thermal gradient engineering.

The present architecture requires a medium optimised for both optical readout — via the magneto-optical Kerr effect (MOKE) — and thermomagnetic writing. Conventional particulate magnetic tape (iron oxide or metal particle coatings in organic binder) is unsuitable for this purpose: the surface roughness scatters incident light, the organic binder degrades under repeated laser heating, and the particulate structure limits domain definition. Instead, the architecture specifies sputtered magneto-optical thin-film coatings — amorphous rare-earth transition metal alloys (e.g., TbFeCo) — deposited on flexible polyester substrates in tape form factor. Such films provide the optical-grade surface smoothness, engineered Curie temperature, large Kerr rotation angle, and inorganic thermal stability required for the combined optical-readout and thermomagnetic-write cycle. Sputtered magnetic coatings on flexible substrates have been demonstrated in data storage research (Sony, Fujifilm), though not to date in magneto-optical configurations optimised for the present application.

2.8 High-Density DNA Information Storage

DNA encodes information at molecular density. Church et al. (2012) demonstrated digital data encoding in synthetic DNA at densities approaching petabytes per gram. The theoretical limit of DNA storage density — approximately 455 exabytes per gram — exceeds any other known physical medium by orders of magnitude.

Jiang et al. (2025) at the Southern University of Science and Technology (SUSTech) have demonstrated a physical DNA cassette tape capable of storing 36 petabytes on a 100-metre film in cassette format. The system uses synthetic DNA strands printed on a plastic tape base, with a barcode indexing system for file-level retrieval and a protective "crystal armour" coating providing projected data stability exceeding 300 years at room temperature. This is a physically realised device, not a theoretical proposal.

The present architecture draws on this development for its context input substrate (Section 3.1.1), exploiting the combination of molecular-density storage, cassette form factor, optically smooth protective coating, and barcode-indexed spatial addressing as the physical basis for petabyte-scale context input via optical imaging.

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3. Proposed Architecture

The architecture is described as a unified system. Its seven components are presented in the order they appear in the inference and training data flow, from physical substrate to weight update and back.

3.1 Physical Substrate Architecture: The Two-Substrate Model

The architecture employs two physically distinct substrate types, each optimised for a different role. This separation reflects a fundamental asymmetry in the system's information flow: context is read-once per inference (write-once media is sufficient), while weights are read continuously and updated iteratively (rewritable media is required).

3.1.1 DNA Context Substrate

The context input substrate provides the raw information against which the neural network performs inference. In the present architecture, this is a DNA cassette tape — specifically, synthetic DNA information printed on a flexible film in cassette format, as demonstrated by Jiang et al. (2025).

The substrate surface is optically imaged rather than sequentially decoded. The illuminated surface is photographed at high resolution, producing a two-dimensional image of the molecular pattern. This image is tiled into segments and provided directly as visual context input to the vision-language inference system.

This approach collapses the conventional storage-to-inference pipeline from six steps — substrate, read head, digital signal, decode, tokens, model — to three: substrate, camera, model. The practical consequence is that the context window becomes physically scalable: more substrate area yields more context; imaging resolution determines context depth per unit area.

At the storage density demonstrated by Jiang et al. (36 petabytes per 100-metre cassette), even partial imaging at achievable optical resolution represents context far exceeding any current digital context window — by many orders of magnitude.

The crystal armour protective coating described by Jiang et al. provides a smooth, durable optical surface suitable for structured illumination. A structured illumination lattice incident on the substrate surface produces a far-field diffraction pattern that is the spatial Fourier transform of the substrate surface pattern — placing the context representation natively in a space mathematically related to token embedding space, and suggesting that a fixed projection layer (one additional SLM) may suffice to map the diffracted output directly into the inference system's embedding space, without conventional vision-language tokenisation.

The barcode indexing system demonstrated in the SUSTech implementation maps directly to the tiling and segment-addressing requirements of the present architecture. Context segments are physically locatable on the substrate via their barcode coordinates, enabling selective illumination of relevant substrate regions — equivalent to retrieval-augmented generation implemented at the physical layer.

DNA context substrates are write-once: the information is fixed at synthesis. This is appropriate for the context role — training data, reference corpora, knowledge bases — where the content is prepared offline and loaded as a physical cartridge. Different cartridges represent different context configurations. Swapping a cartridge changes the system's knowledge base without any digital data transfer.

3.1.2 Magneto-Optical Weight Substrate

The weight substrate stores the current neural network weights as magnetic domain patterns on a rewritable medium. Unlike the context substrate, this medium must support repeated read-write cycles — weights are read at every inference pass and updated at every training step.

The architecture specifies sputtered magneto-optical thin-film tape: an amorphous rare-earth transition metal alloy (typically TbFeCo or GdFeCo) deposited by magnetron sputtering onto a dimensionally stable polyester base film in wide-format tape form factor.

This specification is driven by five requirements that conventional particulate magnetic tape cannot meet:

Optical readout quality. MOKE imaging requires a smooth, reflective surface producing clean polarisation rotation. Sputtered amorphous thin films provide optical-grade surface quality. Particulate tape — magnetic particles in organic binder — scatters incident light and produces noisy, low-contrast Kerr signals.

Thermomagnetic write endurance. The training loop requires repeated laser heating above Curie temperature. Inorganic sputtered films are engineered for thermal cycling, rated at approximately 10^6 rewrite cycles on commercial MO disc media using equivalent alloys. Organic binder in particulate tape decomposes under repeated laser heating — outgassing, local delamination, and progressive degradation of magnetic and optical properties.

Ternary weight encoding. The {-1, 0, +1} weight constraint requires three distinguishable magnetic states readable via MOKE. In sputtered RE-TM films, these map to: domain magnetised in the positive normal direction (+1, positive Kerr rotation), domain magnetised in the negative normal direction (-1, negative Kerr rotation), and demagnetised microdomain region (0, zero net Kerr rotation). The demagnetised state is achieved by heating above the Curie temperature and cooling without applied bias field, producing a random microdomain structure whose net magnetisation — and therefore net Kerr rotation — averages to zero over the optical readout aperture. In continuous amorphous thin films, this averaging is reliable because the microdomain structure is fine relative to the read beam spot size. In particulate media, the discrete particle structure makes the demagnetised state thermally unstable as individual particles drift toward one polarity.

Domain definition. Continuous amorphous thin films support well-defined domain boundaries at sub-micrometre scale, consistent with the spatial resolution required for high-density weight encoding. Particulate media, with randomly oriented discrete particles, limits achievable domain definition and domain wall precision.

Dimensional stability. Sputtered thin films on polyester substrates inherit the base film's dimensional characteristics. While polyester has a thermal expansion coefficient of approximately 17-25 ppm/°C, the film can be tensioned in a fixed-position precision deck with active thermal control to maintain registration within the required tolerances. The inorganic coating does not contribute additional dimensional instability from humidity absorption, unlike particulate coatings with organic binder.

The tape format provides multi-track parallel access: each track corresponds to one network weight layer, addressable by independent optical beams at the track pitch. Professional wide-format tape (2-inch / 50.8mm or wider) provides approximately 24 tracks at standard pitch, accommodating networks of up to 24 layers per tape width. Deeper networks requiring more layers are addressed by wider archival formats, multi-pass addressing, or multiple tape substrates.

This is not a commercial off-the-shelf medium. It requires a custom fabrication run: magnetron sputtering of RE-TM alloy onto flexible polyester substrate, with the coating engineered for the specific Curie temperature, Kerr rotation magnitude, and coercivity required by the architecture. This is established thin-film deposition technology applied to a non-standard substrate geometry — a materials engineering task, not a physics research question.

An alternative near-term path uses commercial MO disc media — which already provides the correct alloy, surface quality, and read-write characteristics — in an automated disc jukebox rather than tape format. This sacrifices multi-track parallel access (discs are single-surface) but eliminates the custom fabrication requirement entirely. Prototyping the optical inference and training loop on commercial MO discs, then migrating to tape format once the architecture is validated, may represent the most tractable experimental sequence.

3.2 Reconfigurable SLM Weight Layers with Ternary Optimisation

Spatial Light Modulator arrays implement the linear weight multiplication layers of the neural network. Each SLM is a grid of individually addressable pixels modulating incident light per pixel; the pixel configuration encodes the weight matrix, and light passing through the configured SLM performs a matrix multiplication in a single optical pass.

The architecture exploits the ternary weight constraint of BitNet b1.58. Weights in {-1, 0, +1} map directly to three discrete SLM pixel states: phase-invert (w = -1), block (w = 0), and transmit (w = +1). This reduces per-pixel configuration from continuous analogue voltage — requiring precise digital-to-analogue conversion and sensitive to fabrication variation — to three discrete states, substantially simplifying both initial configuration and gradient-driven update. SLM layers are electronically reconfigurable on millisecond timescales using current commercial liquid-crystal-on-silicon technology, with ferroelectric LCoS devices offering kilohertz reconfiguration rates and zero static power consumption in hold state.

3.3 Nonlinear Optical Activation

The critical requirement for functional neural network depth is nonlinear activation between linear layers. Without nonlinearity, any cascade of matrix multiplications reduces to a single linear transform. Two candidate implementations are proposed. The viability of passive optical nonlinear activation at practical intensities and across multiple wavelength channels simultaneously is identified as the single most critical experimental validation required by the architecture.

Volumetric metamaterial activation cubes. Three-dimensional photonic crystal structures with geometry engineered at the nanoscale to produce intensity-dependent thresholding via photonic bandgap effects. Below threshold intensity, the periodic structure prevents photon propagation — absorbed or scattered. Above threshold, the nonlinear optical response shifts the effective bandgap, permitting transmission. This is physically equivalent to a rectified linear unit (ReLU) activation function, implemented entirely by structural geometry. The volumetric form factor — a 3D element rather than a flat diffractive surface as in D2NN — provides greater interaction length and therefore stronger nonlinear response at lower intensities. A Fresnel-inspired radially graded internal geometry implements spatially-aware thresholding in which beam centre and beam edges experience different effective threshold conditions. Fabrication via two-photon lithography (direct laser writing) is commercially available at the required sub-100nm resolution.

The metamaterial cube is the preferred long-term solution for free-space integration with the volumetric inference stack. However, it depends on achieving reliable intensity-threshold switching at the optical power levels delivered per channel within the stack. If the required threshold intensities are incompatible with practical compact laser sources, the architecture's core advantage — passive, low-power inference — is compromised. This is an open experimental question. The intensity budget depends on beam diameter, source power per channel, and the specific photonic crystal geometry. Characterising this threshold-intensity budget for the proposed stack geometry is the first experiment the architecture requires.

Fibre-optic saturable absorber sections. Lengths of single-mode optical fibre incorporating graphene, carbon nanotube, or topological insulator saturable absorbers. Fibre confinement — core diameter approximately 8 micrometres over centimetre-to-metre interaction lengths — increases effective intensity by several orders of magnitude relative to free-space beams of equivalent power, making nonlinear switching accessible at practical compact laser powers. Saturable absorbers of this type are commercially deployed in mode-locked fibre lasers; below threshold, incident light is absorbed; above threshold, the absorber bleaches and light transmits. This constitutes a physical ReLU in a commercially available fibre component.

A hybrid architecture — free-space SLM weight layers coupled into short fibre sections for activation, then recollimated — may offer the most tractable near-term experimental path, decoupling the activation proof-of-concept from the more demanding metamaterial fabrication problem. The fibre path adds bulk relative to the volumetric cube form factor and introduces coupling losses at each free-space-to-fibre transition, but it allows the nonlinear activation to be validated using commercial components before investing in custom metamaterial fabrication.

3.4 Wavelength-Division Multiplexed Parallel Inference

N coherent sources at wavelengths lambda_1, lambda_2, ..., lambda_N are combined into a single co-propagating beam traversing the optical stack. At each SLM weight layer, wavelength-selective spatial encoding configures independent weight matrices per channel: either via spatial separation of wavelength channels across SLM regions, via wavelength-selective pixel coatings, or via separate SLMs per channel with beam combination — the simplest engineering approach, compatible with existing commercial hardware. At each activation element, photonic crystal geometry engineered for multi-wavelength response implements independent nonlinear thresholds per channel. At the output face, a diffraction grating or dichroic filter array demultiplexes the N channels for independent detection.

The result is N concurrent, independent neural network forward passes in the same single optical transit time of approximately 0.1-0.15 nanoseconds. The N channels may implement ensemble instances of the same model for majority-vote uncertainty quantification; distinct specialist models applied simultaneously to the same input; pipeline-parallel layers of a model too large for a single stack; or independent queries from distinct input sources. The marginal latency cost of the N-th channel is zero — it traverses the same stack in the same transit time as the first.

Cross-phase modulation between wavelength channels in the nonlinear activation medium is identified as the primary engineering constraint on channel count and isolation. In any medium exhibiting intensity-dependent optical response, the presence of light at one wavelength can modulate the response experienced by light at another wavelength. If this cross-channel interaction is significant, it introduces crosstalk between nominally independent inference channels, degrading inference quality. The severity depends on the specific nonlinear mechanism, the channel spacing, and the total optical power in the medium. If cross-phase modulation limits practical channel count to low single digits, the parallelism advantage reduces from transformative to incremental. This is an open experimental characterisation problem that must be resolved for each candidate activation medium and represents, together with the activation intensity budget (Section 3.3), one of the two experiments that determine whether the architecture's core claims hold.

3.5 Volumetric Cube Array: The 100x100 Inference Substrate

Physical replication of a single WDM optical stack into a planar array, with stacks oriented perpendicular to the array plane, produces a volumetric inference substrate whose concurrent capacity is M x N, where M is the number of stacks and N is the number of wavelength channels per stack.

We specify the concrete architectural instance of a 100x100 planar array. At a beam diameter of 1mm and inter-stack spacing of 3mm, the array occupies a 300mm x 300mm footprint. At an optical path depth of 10-30cm, the complete form factor is approximately 30cm x 30cm x 30cm: a volume of approximately 27 litres, comparable to a large Bluetooth speaker or a small server appliance.

The interior of the cube along the inference axis is structured as follows:

```
[Input illumination plane: 100x100 coherent source array]
        |
[Substrate imaging plane: 100x100 cassette read positions]
        |
[SLM_1: 300mmx300mm aperture, 100x100 independently addressed regions]
        |
[Activation layer_1: metamaterial cube or fibre section array]
        |
[SLM_2 -> Activation_2 -> ... -> SLM_L -> Activation_L]
        |
[Output detector plane: 100x100 detector arrays]
```

The complete optical path traverses 10-30cm of physical depth. At the speed of light in glass (~2x10^8 m/s), the photon transit time from input to output is approximately 0.1-0.15 nanoseconds. This is the actual propagation time — not a theoretical bound — representing the irreducible inference latency of the optical path.

The operating power budget for a 100x100 array at N=10 wavelength channels is estimated at approximately 15-70 kilowatts, dominated by illumination sources and detector readout electronics. This compares to approximately 7 megawatts for an electronic inference cluster of equivalent concurrent capacity at H100-class GPU power draw — a reduction of approximately two orders of magnitude. The metamaterial activation elements and SLMs in hold state draw negligible power; the inference computation is performed by passive photon propagation through physical structures that consume no energy in operation.

The economic consequence is a qualitative inversion of inference cost structure. In an electronic cluster, each additional concurrent inference instance requires additional switching compute drawing proportional power. In the optical cube, additional inference capacity within the existing array is provided by additional wavelength channels — additional laser lines — at marginal cost dominated by the laser source rather than by compute infrastructure. The photons performing the inference are not metered. The structures they traverse are paid for once at fabrication.

3.6 The Photonic Training Loop

The five components described above constitute an inference substrate. The components described in this section close the training loop: enabling the weight substrate to be updated in response to error signals, without digital training infrastructure, using the same optical mechanisms that perform inference.

3.6.1 The Two-Substrate Training Architecture

Training requires two physical substrate roles. The weight substrate (magneto-optical thin-film tape, Section 3.1.2) encodes the current network weights as magnetic domain patterns, with one track corresponding to one network weight layer, addressable by independent optical beams at the track pitch of the tape format. The target substrate encodes the desired output patterns for the current training batch. Both substrates are read by independent optical beams; the weight substrate feeds the inference beam path, while the target substrate feeds the reference beam of the error computation.

For supervised training, the target substrate is a separate magneto-optical tape cassette loaded with the target output patterns for the training dataset. For self-supervised training (Section 7.1), the target signal is generated internally from the system's own inference outputs.

3.6.2 Holographic Error Instantiation

The error signal — the difference between the network's actual output and the desired output — is physically instantiated as the optical interference pattern between two beams at a beam combiner: the output beam from the inference stack and the reference beam from the target substrate. Where these beams are in phase, the inference is correct; where they are out of phase or differ in amplitude, the interference encodes the signed error at each output position.

This is not merely analogous to the holographic process described by Gabor (1969) — it is the same physical process applied to neural network error computation. The interference pattern of two beams is the outer product of their field amplitudes, which is precisely the Hebbian weight update and a component of the backpropagation gradient. The optical interference is performing gradient mathematics. No digital subtraction, no loss function evaluation, no electronic computation is required: the physics of wave interference instantiates the error signal directly.

3.6.3 OAM Gradient Encoding

The interference pattern at the output face represents the error at the output layer. To update weights at earlier layers, the layer-wise gradient tensor must be encoded and directed to the corresponding region of the weight substrate. We propose encoding the gradient for each layer as a superposition of Orbital Angular Momentum modes on a dedicated write beam.

A beam carrying OAM mode l has a helical phase front e^{il *phi}, where phi is the azimuthal angle. Modes at different integer l values are fully orthogonal: the integral of e^{il*phi} * e^{-il'*phi} d_phi = 2*pi*delta_{ll'}. An arbitrary spatial pattern on the beam cross-section — representing the gradient tensor for a given weight layer — can be decomposed into a superposition of Laguerre-Gaussian modes, each characterised by an OAM charge l and a radial index p. The gradient tensor G for layer k is expressed as:

```
G_k(r, phi) = Sum_{l,p} c_{lp} LG_{lp}(r, phi)
```

where c_{lp} are complex expansion coefficients and LG_{lp} are the Laguerre-Gaussian basis functions. A spatial light modulator configured as an OAM mode encoder prepares the write beam in this superposition state. The write beam then carries the full gradient tensor as a structured spatial pattern of phase and amplitude across its cross-section, propagating to the weight substrate at the speed of light.

For ternary-weight networks, the gradient update is coarsely quantised to delta_w in {-1, 0, +1}, substantially reducing the number of OAM modes required to represent the update pattern and simplifying the encoder configuration relative to full-precision gradient encoding.

3.6.4 Thermomagnetic Weight Writing

The gradient-encoded write beam is focused onto the weight substrate at the region corresponding to the target weight layer. The focused spot heats the magneto-optical thin-film locally above its Curie temperature, erasing the previous domain pattern in the heated region. The spatial intensity profile of the write beam — shaped by the OAM

mode superposition to encode the gradient update — determines which sub-regions are heated and to what degree. On cooling under the influence of a bias field or the surrounding domain structure, new domains are written in the pattern prescribed by the gradient-encoded beam profile.

For the ternary weight encoding, three write conditions produce the three weight states:

- **w = +1:** Heat above Curie temperature, cool under positive bias field. Domain magnetised in positive direction. Positive Kerr rotation on readout.
- **w = -1:** Heat above Curie temperature, cool under negative bias field. Domain magnetised in negative direction. Negative Kerr rotation on readout.
- **w = 0:** Heat above Curie temperature, cool without applied bias field. Random microdomain structure forms on cooling, averaging to zero net magnetisation and zero net Kerr rotation over the readout aperture.

The sputtered RE-TM thin film is unconditionally rewritable: the medium retains no memory of previous domain states after heating above the Curie temperature. Professional MO media using equivalent alloys are rated for approximately 10^6 rewrite cycles; the training loop writes new domain patterns at each gradient step without medium degradation within operational cycle counts. The write beam wavelength is chosen to differ from the inference read beam wavelength, with dichroic separation preventing read-during-write interference — a standard technique in magneto-optical recording systems. Ultrashort write pulses enable sub-diffraction-limit domain writing via heat-assisted recording principles, consistent with domain size requirements for ternary-weight representations.

The complete training cycle — inference beam reads weight substrate, output beam interferes with target beam producing error signal, error encoded onto write beam via OAM superposition, write beam updates weight substrate via thermomagnetic writing, next inference pass reads updated weights — is performed without electronic computation in the gradient path. The gradient travels as light. The weight update is written by light. The medium that light writes is the same medium that light subsequently reads for inference.

3.7 Physical Media Handling

Training requires repeated cycling of training data through the target substrate read position. The weight substrate must remain stably positioned throughout. These requirements are addressed by established mechanical infrastructure adapted to the optical substrate architecture.

Weight substrate handling. The magneto-optical weight tape cassette remains loaded in a fixed-position high-precision deck throughout operation, mechanically isolated from the target cycling mechanism. Active thermal control maintains the deck at a stable temperature, limiting dimensional change of the polyester base film. Precision tensioning maintains tape flatness across the read/write aperture. For prototyping on commercial MO disc media, the weight substrate is a disc loaded in a fixed-position precision spindle, with the same mechanical isolation from the target cycling mechanism.

Target substrate cycling. Training target substrate cycling is handled by an auto-loading deck array: a carousel or linear magazine of tape cassettes (or MO discs) from which decks automatically retrieve, thread, and eject cassettes in training batch sequence. This architecture draws directly from robotic tape library systems deployed in enterprise archival infrastructure, which routinely manage hundreds to thousands of cartridges with fully automated retrieval and zero operator intervention.

For homogeneous training across the full 100x100 cube array — all 10,000 stacks training on the same task — a single shared target deck distributes the target beam across the array via beam-splitting optics, reducing the auto-loading mechanism to a single high-throughput unit. For heterogeneous training tasks, independent target decks per stack or stack group provide task isolation.

DNA context substrate handling. Context cassettes are loaded into a separate read-only deck array. Context cassettes are not modified during operation — they are loaded at the beginning of a task and remain in position for the duration. Changing the system's knowledge base is a physical operation: eject one cassette, insert another. The auto-loading mechanism for context cassettes draws on the same robotic library infrastructure as the target deck.

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4. Novel Contributions

The following contributions are identified as not present in the prior art as an integrated system. Individual component concepts are explicitly acknowledged as drawing upon the prior work cited in Section 2; the novel claim is their specific combination:

- **Two-substrate physical architecture with role separation.** The specification of a write-once DNA context substrate (petabyte-scale, optically imaged) and a rewritable magneto-optical weight substrate (thermomagnetically updated, MOKE-readable) serving complementary roles in a unified inference and training system. The context substrate provides the data; the weight substrate provides the learned parameters. Different media, different form factors, different decks.
- **Physical substrate optical imaging as context input modality.** The treatment of magnetic or molecular substrate surface patterns as direct visual context input for an inference system, bypassing sequential digital decoding, with context window scalability determined by substrate area and imaging resolution, and with structured illumination lattice producing Fourier-domain output natively related to embedding space.
- **Ternary-optimised SLM weight layers.** The explicit mapping of BitNet b1.58's {-1, 0, +1} weight constraint to three discrete SLM pixel states — phase-invert, block, transmit — reducing optical weight layer configuration from continuous analogue to three discrete states and substantially simplifying both inference operation and gradient encoding for training.
- **Ternary magneto-optical weight encoding.** The mapping of three-valued neural network weights to three distinguishable MOKE readout states in sputtered RE-TM thin films: positive domain (+1), negative domain (-1), and demagnetised microdomain average (0), with the zero state achieved by field-free cooling after thermomagnetic erasure.
- **Volumetric metamaterial activation cubes and fibre-optic saturable absorber sections as passive nonlinear activation.** The specification of both a 3D photonic crystal volume element and a fibre saturable absorber section as passive intensity-threshold activation functions, with the volumetric geometry providing interaction-length advantage over flat diffractive surfaces and the fibre geometry providing practical power-level advantage over bulk free-space nonlinear elements.
- **Wavelength-division multiplexed parallel neural network inference.** The application of WDM channel independence to N concurrent independent model forward passes through a single physical optical stack, with wavelength-selective SLM encoding and photonic crystal activation implementing independent per-channel weight matrices and activation thresholds.
- **Volumetric 100x100 cube array as a deployable inference substrate.** The specification of a three-dimensional array of optical inference stacks in a 30cm cube form factor, providing 10,000xN concurrent frontier-class model inferences at kilowatt-scale power with no processor, no memory bus, no clock, and no network interface.
- **OAM-encoded gradient beam for layer-wise photonic weight update.** The decomposition of neural network gradient tensors into Laguerre-Gaussian OAM mode superpositions and their encoding onto a laser write beam, enabling layer-specific gradient information to propagate to the weight substrate as structured light without digital conversion.
- **Holographic error instantiation via output-target beam interference.** The physical computation of the neural network error signal as the optical interference pattern between the inference output beam and a target reference beam, directly implementing the outer-product gradient component as a holographic process without electronic subtraction or loss function evaluation.
- **Thermomagnetic domain writing as the photonic weight update mechanism.** The use of a gradient-encoded focused laser write beam to perform thermomagnetic weight updates on a rewritable magneto-optical thin-film substrate, closing the training loop without electronic write heads or digital-to-analogue conversion, on medium rated for approximately 10^6 rewrite cycles.
- **Multi-track wide-format tape and auto-loading deck array as the physical training infrastructure.** The specification of professional wide-format magneto-optical tape as a parallel multi-layer weight substrate and the application of robotic tape library mechanisms to training target substrate cycling, providing continuous automated training operation without operator intervention.

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5. Applications

5.1 Sovereign Deployable Intelligence

The combination of cube array inference, physical substrate weight storage, thermomagnetic on-device training, and zero required network connectivity describes a deployment class with no precedent in electronic AI infrastructure: a self-contained, physically portable intelligence substrate running and training tens of thousands of concurrent frontier-class model instances at kilowatt-scale power with zero digital attack surface when isolated.

Model weights encoded as magnetic domain patterns on removable MO tape cassettes have no digital attack surface when the cassette is physically removed from the system. They cannot be remotely accessed, exfiltrated via network intrusion, or tampered with through software vulnerabilities. Physical custody of the cassette is physical custody of the model. A 30cm cube drawing 15-70kW is deployable in a vehicle, a secure room, or a field installation with generator or solar-plus-battery power. The electronic equivalent — a 10,000-GPU cluster — requires a purpose-built building, a dedicated grid substation, and continuous physical and cybersecurity operations.

Direct applications include: air-gapped intelligence analysis environments; sovereign edge inference for defence and national security deployments; physically authenticated model distribution via cassette media with tamper-evident custody chains; and field-deployed autonomous systems requiring frontier-class intelligence without communications infrastructure.

5.2 Ensemble and Uncertainty-Quantified Inference

N concurrent instances of the same model on the same input — achieved at zero latency penalty via WDM — enables ensemble inference and calibrated uncertainty quantification as a native architectural property. Outputs that diverge across wavelength channels flag low-confidence inferences for human review. Majority voting across channels provides robust outputs resistant to individual-channel noise. This is practically valuable in high-stakes inference domains — medical, legal, financial, security — where single-model output confidence is operationally insufficient.

5.3 Parallel Specialist Inference

Different wavelength channels implement different models simultaneously: a reasoning specialist, a code specialist, a factual retrieval specialist, a safety classifier, all applied to the same input in a single optical transit. The combined output of specialist models, demultiplexed at the detector plane, provides a richer inference product than any single model at equivalent latency. This is a physically natural implementation of mixture-of-experts architectures, realised at the optical physics level rather than in software routing logic.

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6. Limitations and Open Problems

The architecture is pre-experimental. The following represent significant open engineering and physics problems whose resolution is required before experimental validation of the integrated system. They are presented in order of assessed criticality to the architecture's core claims.

6.1 Critical Path: Nonlinear Activation at Practical Intensities

This is the single experiment that determines whether the architecture works. Photonic crystal bandgap-mediated switching may require intensities incompatible with compact low-power sources. The fibre saturable absorber alternative addresses this for near-term experiments but adds bulk and coupling losses. The metamaterial cube remains the preferred long-term solution for free-space integration and requires further materials engineering. If neither approach achieves reliable activation at the optical power budget of the proposed stack, the architecture does not produce functional neural network inference regardless of the performance of every other component. Characterising the activation intensity budget for the proposed geometry is therefore the first-priority experiment.

6.2 Critical Path: Cross-Phase Modulation and Channel Isolation

The second experiment that determines whether the architecture's parallelism claims hold. Cross-phase modulation between WDM channels in the nonlinear activation medium could impose a hard ceiling on practical channel count. If the ceiling is low (3-4 channels), the parallelism advantage reduces from transformative to incremental. The severity is medium-specific and must be characterised for each candidate activation element.

6.3 Signal Attenuation Across Stack Depth

Light intensity decreases with each optical element traversed. Without gain, signal-to-noise ratio degrades across network depth, limiting the practical number of layers. Nonlinear optical gain materials — providing optical amplification between layers without electronic conversion — are a potential solution and represent an active research area. However, gain adds noise, power consumption, and complexity. The attenuation budget per layer, the minimum detectable signal at the output plane, and the maximum achievable depth without gain must be characterised experimentally. This may prove to be the fundamental limit on network depth in the optical domain, analogous to vanishing gradients in electronic deep learning but imposed by physics rather than mathematics.

6.4 Magneto-Optical Weight Substrate Fabrication

The architecture requires sputtered RE-TM thin-film coatings on flexible polyester substrates — a non-standard substrate geometry for magneto-optical materials. While the individual technologies (magnetron sputtering, RE-TM alloys, flexible substrate coating) are established, their combination for this specific application has not been demonstrated. Coating uniformity, adhesion, and magnetic property consistency across wide-format tape lengths are open fabrication questions. The MO disc prototyping path (Section 3.1.2) provides a near-term workaround using commercial media.

6.5 Large-Area SLM Fabrication

The 300mm x 300mm SLM aperture required for a 100x100 array exceeds current commercial single-element formats. Tiled SLM arrays and large-area SLM development programmes are candidate engineering paths.

6.6 Thermal Management of the Illumination Plane

10,000 laser sources in a 300mm x 300mm plane generate heat causing differential phase errors. Active thermal management or athermal optical design is required.

6.7 OAM Mode Crosstalk and Gradient Encoding Fidelity

Optical aberrations and substrate irregularities degrade OAM mode orthogonality, introducing gradient encoding errors. Adaptive optics compensation adds complexity and power cost.

6.8 Thermomagnetic Write Resolution

Thermal diffusion limits domain boundary definition beyond the optical spot size. HAMR techniques partially address this; adaptation to the MO tape substrate formats proposed requires experimental characterisation.

6.9 Backpropagation Through Activation Elements

Computing gradients at weight layers below the output requires propagating error through the nonlinear activation elements. Phase conjugation and in-situ measurement approaches (following Hughes et al., 2019) are candidate mechanisms; neither is straightforwardly applicable to the magneto-optical substrate write architecture without further development.

6.10 Detector Data Rate at Array Scale

10,000xN concurrent detector channels at practical frame rates imply aggregate readout data rates potentially exceeding 100 Tbps, requiring custom high-bandwidth readout architectures.

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7. Open Questions: Toward Autonomous Self-Directed Learning

The training loop described in Section 3.6 operates in a supervised setting: the target substrate encodes desired outputs, and the error signal is the optical difference between actual and desired. The system trains on-device and without external digital infrastructure, but the training direction — what it is learning to do — is specified externally, encoded on the target substrate. Two open questions are identified whose resolution would determine whether the architecture supports fully autonomous self-directed learning.

7.1 The Self-Generated Target Signal

In unsupervised and self-supervised learning, the training target emerges from the structure of the data itself rather than from external labels. The question is whether such targets can be generated and encoded photonically, without an external digital pipeline, from the system's own inference outputs.

A specific candidate mechanism is predictive coding (Rao & Ballard, 1999): each layer predicts the activity of the layer below, and the prediction error is the training signal. In the optical domain, the prediction at layer k is the output of SLM_k applied to the activation of layer k-1. The error is the interference between this prediction beam and the actual layer k-1 activity beam. Both beams are already present in the inference stack. No external target substrate is required; the target reference beam is derived from the system's own internal optical activity.

Whether this constitutes a practical unsupervised photonic training mechanism is an open experimental question, but the required beams are physically present and the interference mechanism is the same as in the supervised case.

7.2 The Autonomous Direction Question

A system that generates its own training signal from its own inference outputs, cycles its own target media through an auto-loading deck, and writes its own weight updates via thermomagnetic beam is, in a meaningful physical sense, directing its own learning. The training trajectory is determined by the system's own inference dynamics rather than by external specification of a loss function.

Whether this is desirable, safe, or interpretable in any given deployment context is outside the scope of this architectural disclosure. It is noted here because it is the logical terminus of the architecture: the point at which the sovereignty argument is complete not merely in infrastructure — no external compute required — but in direction. These two questions are left explicitly open. They define the primary frontier of investigation extending from the present work.

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8. Conclusion

We have described a unified architecture for neural network computation in which inference, massive parallelism, and on-device training are performed primarily in the optical domain. The architecture proposes no new physics: every mechanism invoked — optical matrix multiplication, WDM channel independence, OAM mode orthogonality, holographic interference, thermomagnetic domain writing, DNA molecular-density storage — is established in the relevant literature. The contribution is their combination into a coherent system with properties not achievable by any individual component or prior combination.

The system that emerges from this combination is a 30-centimetre cube drawing kilowatt-scale power, running tens of thousands of concurrent frontier-class neural network instances, training itself on physical media that light writes and rewrites, with no processor in the inference path, no memory bus, no clock, and no required network connection. Context is loaded as DNA cassette cartridges carrying petabytes of information at molecular density. Weights are stored as magnetic domain patterns on magneto-optical tape, readable by polarised light and writable by focused laser. The electronic equivalent of its inference capacity requires a purpose-built data centre drawing megawatts. The photonic architecture draws the power of a small workshop and fits in a vehicle.

The engineering challenges are substantial and are not minimised here. Every component of the integrated system requires experimental validation. The path from this disclosure to a working device is long and uncertain. Two experiments are identified as critical: nonlinear optical activation at practical intensity budgets, and cross-phase modulation characterisation for WDM channel isolation. If either proves fundamentally limiting, the architecture does not achieve its core claims. But there are no known fundamental barriers in the relevant literature — only engineering problems of varying difficulty, each with candidate solutions in established research.

What remains open, and is deliberately left open, is whether the system can close the final loop: generating its own training signal from its own outputs, directing its own learning without external specification. That is the question this architecture makes physically meaningful to ask.

This document establishes the complete unified architectural conception as prior art as of 13 March 2026. The authors invite experimental collaboration and welcome correspondence from researchers in photonic computing, metamaterial optics, magneto-optical recording, large-scale optical fabrication, OAM communications, DNA information storage, and AI inference systems.

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